Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/16/2016
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sgovind@ccny.cuny.edu	**Fax:**	FY 212-650-8585
PI Organization Type:	UNIVERSITY	**Phone:**	212-650-8476
Organization Name:	City College of New York		
PI Address 1:	Biology Department		
PI Address 2:	160 Convent Ave		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of "space-cultured" wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. We have analyzed the proteome of virus-like particles (VLPs) and found no evidence for a viral nature of VLPs. Instead VLPs appears to possess properties of eukaryotic microvesicles. Curiously, they also possess proteins with prokaryotic character. To our knowledge, this is the first kind of an virulence-linked organelle-like structure produced in eukaryotes with a mixed prokaryotic/eukaryotic character. 2. We have characterized the lethal effects of VLPs on the hematopoietic organ of fly larvae called the lymph gland. This organ houses stem-like progenitors which are targeted by VLPs for destruction. 3. We have discovered the presence of bioactive lipids in fly larvae and have monitored their levels in a Drosophila model of tumor-inflammation. Aspirin administration relieves inflammation in flies and modulates the levels of the bioactive lipids. All these studies open new areas of discovery and strengthen the Drosophila system as a stand-in for human and animal studies in space.
Task Progress:	The Principal Investigator and Co-Investigator's labs are ready to adapt and optimize the Vented Fly Box (VFB) prototype according to the experimental plan proposed in the grant and the Experiment Requirements Document (ERD). Three presentations were made in conferences and two manuscripts are being prepared for publication.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Heavner ME, Ramroop J, Gueguen G, Ramrattan G, Dolios G, Scarpati M, Qiu W, Wang R, Singh S, Govind S. "Molecular analyses of immune-suppressive virus-like particles from a Drosophila parasitic wasp suggest cell-specific activities and a hybrid biotic particle nature." Oral Presentation, The Allied Genetics Conference 2016, Genetics Society of America, Orlando, FL, July 13-17, 2016. Program and Abstracts. The Allied Genetics Conference 2016, Genetics Society of America, Orlando, FL, July 13-17, 2016. PgmNr D228, p. 86-17. , Jul-2016
Abstracts for Journals and Proceedings	Heavner ME, Ramroop J, Ramrattan G, Dolios G, Crissman J, Scarpati M, Kwiat J, Miller L, Qiu W-G, Wang R, Singh S, Govind S. "Multi-'omics' and structural biology approaches identify a novel vesicle-based secretion system in Drosophila parasites." Poster Presentation, The NY Structural Biology Discussion Group Summer 2016 Meeting, New York, NY, August 4, 2016. The NY Structural Biology Discussion Group Summer 2016 Meeting, New York, NY, August 4, 2016. , Aug-2016
Articles in Peer-reviewed Journals	Lin CJ, Smibert P, Zhao X, Hu JF, Ramroop J, Kellner SM, Benton MA, Govind S, Dedon PC, Sternglanz R, Lai EC. "An extensive allelic series of Drosophila kae1 mutants reveals diverse and tissue-specific requirements for t6A biogenesis." RNA. 2015 Dec;21(12):2103-18. Epub 2015 Oct 29. http://dx.doi.org/10.1261/rna.053934.115 ; PubMed PMID: 26516084; PubMed Central PMCID: PMC4647464 , Dec-2015
Articles in Peer-reviewed Journals	Gokhman VE, Bolsheva NL, Govind S, Muravenko OV. "A comparative cytogenetic study of Drosophila parasitoids (Hymenoptera, Figitidae) using DNA-binding fluorochromes and FISH with 45S rDNA probe." Genetica. 2016 Jun;144(3):335-9. http://dx.doi.org/10.1007/s10709-016-9902-5 ; PubMed PMID: 27150102 , Jun-2016